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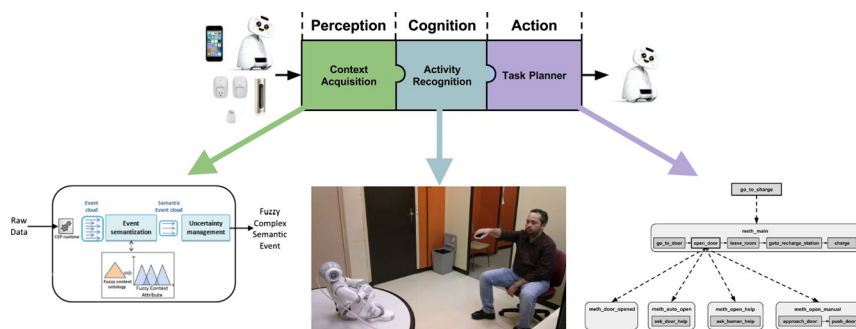
A Framework for Service Robots in Smart Home: An Efficient Solution for Domestic Healthcare

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HIGHLIGHTS

- Proposition of a complete framework for healthcare service robot in smart homes.
- Division in three steps: perception, cognition and action.
- Challenges identified and solutions proposed for each layer.
- Each solution's efficiency is assessed through specific experiments.

GRAPHICAL ABSTRACT



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ABSTRACT

Purpose: As the elder population grows, the need for domestic healthcare is on the rise. Both robotics and smart environments, including smart homes, provide a promising solution to monitor, interact and keep company to users. However, in real case scenarios, sensors data are not perfect and the environment changes over time, leading to erroneous understanding of the context and inappropriate responses. The purpose of this work is to tackle those challenges in order to improve the autonomy and efficiency of robots in smart environments.

Methods: The problematic was structured into three steps: (1) perception, (2) cognition and (3) action. We proposed and evaluated a software framework that covers the challenges of each step. It includes respectively: (1) a context acquisition method that supports and models the uncertainty of data by using complex event processing, fuzzy logic and ontologies; (2) an activity recognition system that combines vision, context knowledge and semantic reasoning; (3) a dynamic hierarchical task planner that alternates planning and execution. For each step, the framework was evaluated through simulations and/or experiments using a robot and a smart room.

Results: The quality of the perception was assessed by measuring the efficiency of a cognition process using the acquired context knowledge. An uncertain environment was simulated, and results show our framework to enable a gain of 10% of correctness for an activity recognition process. The cognition part of the framework was evaluated by observing several persons performing activities. It achieved an overall 90% correct recognition, yet, such result questions the relevance of our approach. Finally, the action step was confronted a simulated scenario with various levels of dynamism. Our task planner appeared to reduce, by up to 23%, the number of tasks required to reach a goal in a dynamic environment.

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Conclusion: Our framework provides software tools that make robots and smart environments more relevant in real housings. By supporting the uncertainty of context data and the dynamism of the environment, robots and smart environments can achieve more effectively their purposes in domestic healthcare applications.

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1. Introduction

As the population is growing older, the population subjected to isolation and health issues is on the rise. This creates a rising need for domestic healthcare. In response, we currently observe the emergence of both service robotics and smart environments at home. While different, robots and smart environments, including smart homes, are increasingly used together, as depicted in projects such as Robot-Era [1] or CompanionAble [2]. In fact, by combining the mobility and capabilities of robots, and the range and variety of a smart environment, a high quality and autonomous service can be provided to the user. Such a system can efficiently monitor the activity of the user, understand situations from different sources of data, intervene through actuators or robots, interact with a person, maintain a company to the user, and/or alert the medical staff if necessary.

Although promising, current solutions are however facing the constraints of the reality. Indeed, in real conditions, environments are far from being perfect for robots and smart devices. Typically, sensors can be mistaken and provide uncertain data, for example, observations from robots and devices may be in contradiction. This leads to an erroneous recognition of the user activity or of the situation. Similarly, as the user perform various actions daily, the environment is highly variable, causing interventions of the system to be altered and to potentially fail. These lead to the questions: how can robots and smart environments understand the context, including user activity, from uncertain data? How can they intervene in a variable environment?

Handling uncertainty for activity recognition or situation identification has been explored on several occurrences in the literature [3–5]. However, existing solutions only partially tackle the issue of uncertainty by focusing on specific dimensions, such as the imprecision [5] or freshness [6]. Yet, in order to be compliant with real case scenarios, robot and smart environment should handle all uncertainty aspects. To intervene on the environment, robots, helped by smart devices, can rely on task planner. However, as the environment varies, the computed plan needs to be adapted to prevent task failures. This is mostly performed by recreating the plan after a failure [7]. By not proactively avoiding the failures, the robot wastes time and energy in futile task executions. In the literature, even if some solutions control the replanning phase [7,8], no propositions explore the possibility of dynamically adapting the plan.

In this paper, we present a set of methods to tackle these challenges over the perception, cognition and action of a robot in a smart environment, thus improving their efficiency for healthcare applications.

2. Material and methods

2.1. General framework

In this section, we describe the global structure of our framework. It relies on three steps, namely perception, cognition and action:

- **Perception:** the robot gathers context data through various sources and creates a context knowledge as an ontology.

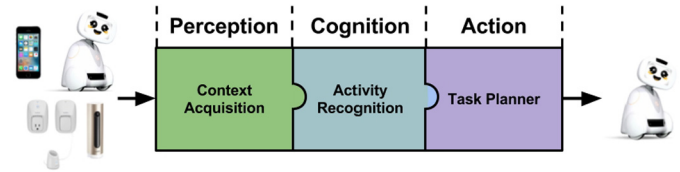


Fig. 1. Three-steps architecture of the framework.

- **Cognition:** the robot analyzes the context knowledge and reasons to understand what is happening in the environment leading to a decision.
- **Action:** the robot acts on the environment by creating a task plan from the knowledge of the context and a goal determined by a cognition decision.

The global process is described in Fig. 1. For each step, we provide one or multiple tools that solve particular challenges. We will now identify the challenges for each step.

2.1.1. Perception

The first step of our system is to acquire context data from various sensors, from both the robot and/or the smart environment. In domestic environment, uncertainty is a serious issue. In fact, we should expect that sensors will provide non-perfect data. For instance, a thermometer may not be accurate or a motion sensor may be intempestively triggered, leading to uncertain context data. Subsequently, this may cause problems or miscomputations in the cognition layer. For instance, if a sensor implies that the user is in room A while he is in room B, a wrong activity may be recognized. As expressed by Ye et al. [3], there are multiple dimensions of uncertainty. In a context of a robot operating in a smart home, we identified four essential dimensions:

- **Freshness:** data is outdated.
- **Precision:** data is correct yet inexact.
- **Accuracy:** data is wrong or partially wrong.
- **Contradiction:** two pieces of data provide contradictory information.

Thus, we need a tool that is able to tackle these uncertainty dimension and model context data accordingly in order to be compliant with the cognition layer. The proposed solution is described in Section 2.2.1.

2.1.2. Cognition

The cognition step aims to analyze the context to understand what is happening and take a decision accordingly. For example, if the robot understands the user has fallen down by reviewing its context knowledge, it should warn the medical staff. The cognition consists in recognizing an activity and reacting accordingly. Activity recognition aims to identify what the user is doing, willingly or not. 'Cooking', 'Sleeping' and 'Falling' are possible activities. The robot is able to recognize what the user is doing from vision [9, 10]. However, these image based techniques can be a bit inaccurate and require a lot of training. In fact, the main limitation is the confusion between activities: some activities imply doing similar gestures and it sometimes can be hard to distinguish them. Fig. 2

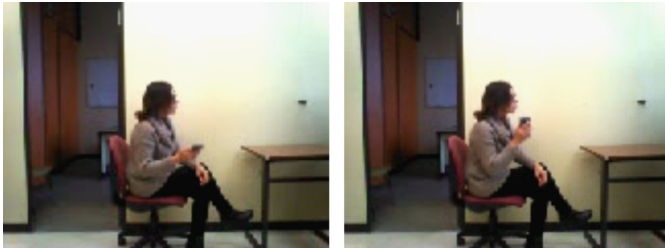


Fig. 2. Video sample that illustrates a confusion between activities (*drinking or phoning*).

shows such an example. But, in our context, the robot camera is obviously not the only sensor and various context data can help the activity recognition process. We propose an enhancement to a vision based activity recognition process that adjusts the recognition's result according to the context knowledge.

Proposed solution for cognition is presented in Section 2.2.2.

2.1.3. Action

The action layer aims to intervene in the environment in order to reach a goal that was set by the cognition layer. To do so, it relies on a task planner: a tool to generate a sequence of tasks to perform in order to reach a goal. Automated planning is a common subject and numerous solutions exist, in particular STRIPS (Stanford Research Institute Problem Solver) [11] or HTN (Hierarchical Task Network) [12–14] are pioneer works. A classical task planner uses as an input the context and a goal, and outputs a sequence of actions, that is the plan. The task planner is essential for domestic robot as it allows them to find a solution according to the context which can vary in homes. Moreover, these environments are also dynamic as other actors, in particular users, continuously alter them. This variability is a problem for plan execution as changes in the context can make a plan outdated and irrelevant. For example, if the robot is asked to fetch some medications and the user moves them while the robot is reaching their location, it won't be able to find them, thus failing to achieve its goal. Another issue in smart homes is the availability and accessibility of the data during planning. For example, without dedicated sensor, the robot cannot be sure if a door is open or closed until it can see it. It is important to take these two aspects into consideration for the task planner in order for the robot to provide a reliable care. To solve these two challenges, we propose an upgrade of the HTN planner that combines planning and execution to enable flexibility and reactivity to the robot's plan. It is addressed in section 2.2.3.

2.2. Contributions

As expressed in the previous section, each step has specific challenges for our context of a robot operating in a smart home. Consequently, specific solutions must be proposed to each of these challenges in order to provide an efficient framework.

2.2.1. Perception

As expressed in Section 3.1, context acquisition is facing four types of uncertain data: freshness, accuracy, imprecision and contradiction. We propose a tool that combines Complex Event Processing (CEP), semantic knowledge and fuzzy logic. Our solution is a three steps process, depicted in Fig. 3.

1. **Event processing:** raw events acquisition from sensors, batching and filtering. Relies on a CEP.
2. **Semantization:** enhancing events with semantic knowledge and format.
3. **Fuzzification:** computation of fuzzy values by analyzing batches of events and generation of one Fuzzy Semantic Complex Event (FSCE).

The resulted FSCE is sent to the cognition layer tools that will store and use them. We will now explain each step of the context acquisition process and how they tackle the uncertainty. For a more detailed explanation, please refer to [15].

Event processing. The aim of the event processing step is to gather events from various sources filter them on given criteria and batch the event in time-windows. It relies on a classical CEP system such as ESPER [16]. A CEP is a tool that gathers event and generates higher level ones. In our work, we use it for collecting data while the computation of higher level event (FSCE) is done by the next two steps. The CEP relies on designer-defined query such as the following one: *Select events of type='location' in time_batch(1min) where source='motion1' or source='camera1'*. This query gathers the event of type 'location', filters by their sources and batch them in 1 min time-windows. The filtering allows to do a first basic handling on events. The time batching allows to consider only the events in a given time window, hence we prevent using outdated event, thus tackling freshness. This step outputs a batch of filtered events, which is provided to the semantization process.

Semantization. Based on a background knowledge represented as an ontology, the semantization phase enriches and formats the events provided in the batch. The enrichment has two purposes. First, it tackles the imprecision issue by enhancing event data through further information stored in the expert-provided background. For example, a motion sensor can be linked to a given room and a particular user. Based on this knowledge, the semantization phase will enrich this sensor's events so that they can carry the information of the user location. The second purpose of the enrichment is to cope with the accuracy of sensor by associating them with a confidence value provided by an expert. This confidence value aims to represent and tackle the accuracy of sensor. In order to carry these further data, the events are formatted into RDF (Resource Description Framework) graphs. The resulting events carry enriched information presented as object linked to their properties, as depicted in Fig. 4. The structure of these graphs is computed from the background knowledge. By formatting the

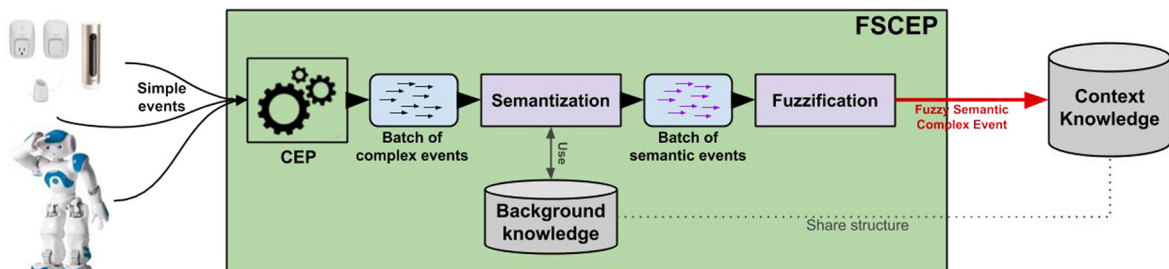


Fig. 3. Context acquisition process.

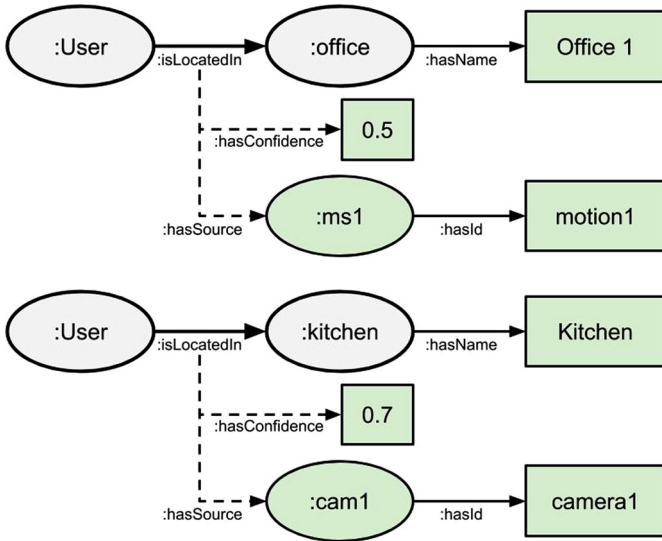


Fig. 4. Example of events formatted as RDF graphs.

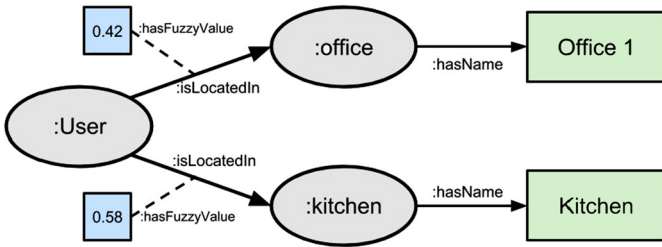


Fig. 5. Example of a $FSCE_{location}$.

event as RDF graph, the data are interoperable and can be easily integrated to other system, in particular cognition's tools in our case. The semantization outputs a batch of semantic events that is passed to the fuzzification phase.

Fuzzification. The fuzzification phase analyses a batch of semantic events to generate one Fuzzy Semantic Complex Event (FSCE), that is an event that embeds multiple values linked to weights. For example, as for location, the fuzzification can generate such an event: $FSCE_{location} = \{[office; 0.42], [kitchen; 0.58]\}$. This allows our system to overcome contradiction: it provides a trust value for each value and it does not exclude any possible value, meaning all possibilities will be taken into account in the cognition layer. To compute trust weights, the process uses a dedicated membership function on all events of the provided batch. It uses the semantic information carried by events. In our case, we use a weighted mean on confidence value, but other functions can be used the designers. The process then generates one single FSCE as a RDF graph with the computed trust weights modeled as annotation, as depicted in Fig. 5. The generated FSCE is then stored in an ontology that is used by the cognition layer.

2.2.2. Cognition

Once the context knowledge is acquired and stored in an ontology, the cognition step comes in play. It is composed by an enhanced vision based activity recognition system that leads to a simple decision making process.

The robot has the ability to recognize the activities of the user. However, as expressed in Section 2.1.2, it can encounter difficulties. We enhance the vision process presented in [9,10]. This approach relies on a four-steps process to output a list of activities associ-

ated with a probability, for example: $\{(eating, 0.05), (phoning, 0.1), (walking, 0.01, \dots)\}$. It was proven efficient on benchmarks, but achieve limited results in real case scenario, such as ours. To do so, we enrich the vision process with the available context knowledge. First, an mentioned earlier the context knowledge is stored and maintained in an ontology that is filled from the perception step. In other words, the ontology carries data from the FSCE, including the trust weights. Whenever the robot applies its vision algorithm for activity recognition, it checks the context knowledge and enhances its results based on simple rules. Let us have an example: "If the phone's inertial unit exceeds a given threshold-increase 'phoning' probability".

The increase of the probability is due to trust weights of data, but the way of activities probabilities being increased is defined by the designer. Such adjustments lead to a more accurate activity recognition. However the background knowledge can also refine the result of the vision algorithm. In fact, it is possible to detect more precise activities by combining the vision results and the context data. For example, if the user is 'sitting' in the kitchen around noon, the robot can infer that he/she is eating. To do so, we apply reasoning rules. Here is an example written using Jena: $(?act \text{ is-a } Activity) (?act \text{ a label "sitting"}) (?act \text{ a proba } ?prob) (?any-act \text{ a proba } ?anyprob) \text{ greaterThan } (?prob \text{ anyprob}) (?usr \text{ located-in } ?room) (?room \text{ is-a } RoomMeal) (?meal \text{ time meal } ?time) \text{ equal } ("current \text{ time" } ?time) \rightarrow (?newAct \text{ is-a } Activity) (?newAct \text{ a label "eating"})$. By enhancing the vision-based activity recognition with semantic background knowledge, we improve the accuracy of the results as well as their precision. We conducted multiple experiments that are addressed in Section 5. The output is the same as the vision process, namely a list of activity associated with refined probabilities. These activities are then used for decision making [17], that determines a goal for the robot to achieve.

2.2.3. Action

Once it has taken its decision, the robot has a goal to achieve. It now has to determine how to achieve the goal. To do so it relies on a task planner. In our work, we selected and improved HTN (Hierarchical Task Network).

HTN is a common and mature planning technique. It considers two types of tasks: primitive and compound. The algorithm consists in decomposing compound task into sub-plan, called methods, according to preconditions and the current context. It finishes when all remaining tasks are primitive. An example a HTN for battery recharging scenario can be found in Fig. 6. For the complete formalism, please refer to dedicated works [13,12,18,19]. By using the knowledge of methods, HTN offers a smaller research space as well as a better control on these generated plans, for these reasons, it was widely used in robotics [20,21,7], including our work.

Nevertheless, HTN is subjected to the same challenges as any task planner: in our case, the dynamism of the environment and the unavailability of some data when planning. To tackle these problems, we propose an extension of HTN, Dynamic HTN (DHTN) [22], featuring a combination of planning and execution: instead of generating the whole plan at once, DHTN incrementally builds the plan while executing it. By doing so, DHTN is able to take into account fresh context data and proposes up-to-date plans. DHTN mainly uses the same formalism as HTN as a base and uses the context knowledge acquired during the perception step. But as DHTN also handles the execution of the plan, the notion of status was added to the tasks, operators and methods. There are three states: SUCCESS, FAILURE and RUNNING. DHTN's elements are associated with these statuses as the plan is executed and the performed tasks fails or succeeds. This enables DHTN to monitor the plan execution and to use this information for planning. For instance, statuses are particularly useful to find an alternative so-

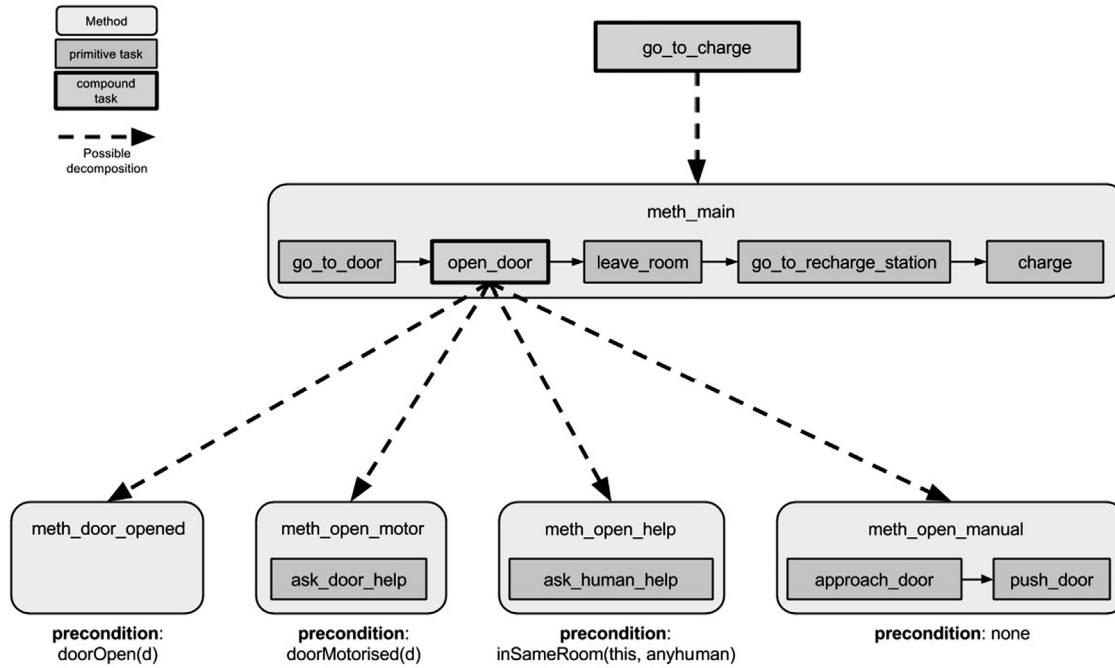


Fig. 6. Simple example of a HTN.

lution. DHTN relies on a novel algorithm. Starting from an initial task, DHTN operates as follows:

- If the current task is a primitive task: DHTN runs it and the robot executes it. DHTN waits for the task's result. If the task succeeds (status SUCCESS), DHTN moves to the next task and reapplies the algorithm. If the task fails (status FAILURE), its parent method and compound task also fails. DHTN backtracks to the parent compound task and tries to find another method. If there is no parent task, DHTN fails to achieve the goal.
- If the current task is a compound task: DHTN compares the current task's methods' preconditions to the current context and selects a compatible method. By doing so, DHTN takes into account the last changes in the environment and selects an up-to-date sub-plan. Furthermore, by selecting a method when required, the robot can access data it could have missed if it would have planned beforehand. DHTN then moves to the first subtask of the selected method and reapplies the algorithm

By combining execution and planning, DHTN is able to manage a reactive and flexible plan. It limits the number of failures and, in case the failure could not be prevented, finds alternative solution without launching the whole process again.

2.3. Materials and technologies used

Our framework was implemented and tested through both simulation and physical tests. Implementations were realized in Java for the perception and cognition, and in C++ and Python for the action phase. For simulation, we rely on Freedomotic (<http://www.freedomotic.com/>), an open source development framework to manage smart spaces, in particular smart homes. Freedomotic allows to define data sources, actuators and carries a simple event processing system. The represented devices can be linked with real ones or simulated ones: by doing so, it is easy to implement in real life a system tested in the framework. One of Freedomotic main feature is its compliance with plug-in: we can easily find any plug-in we might need thanks to a play-store like tool and we can

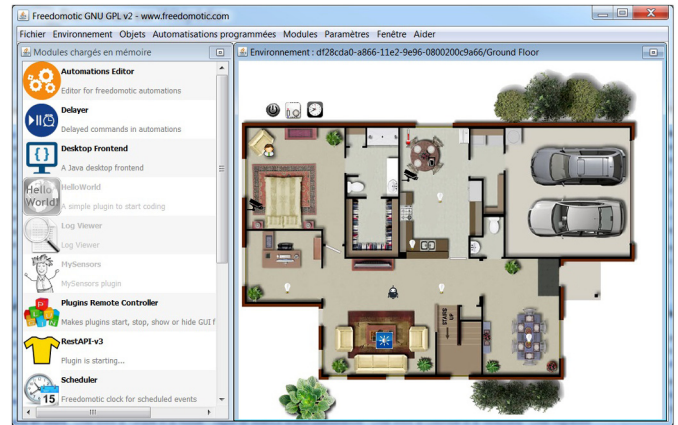


Fig. 7. Freedomotic interface.

create our own upgrades. Fig. 7 shows a screenshot of Freedomotic interface.

Furthermore, we also tested our contributions with a robot and a smart environment. We used a Nao H25 robot [23], a popular humanoid robot that has a lot of abilities, such as locomotion and speech. ROS [24] was used in order to control the robot and interact with our framework. As for the smart environment, we relied on the Hadaptic (<http://hadaptic.telecom-sudparis.eu/fr/introduction/>) platform. The platform is composed by a modular room, furniture and various sensors, such as motion sensors, opening sensors, thermometers and beacons. They are depicted in Fig. 8.

3. Results

Each layer was tested independently. Thanks to it, we aim to evaluate the strengths and weaknesses of each contribution.

3.1. Perception

Our context acquisition approach was tested through a simulation environment based on Freedomotic. We evaluated the



Fig. 8. Nao robot and Hadaptic platform.

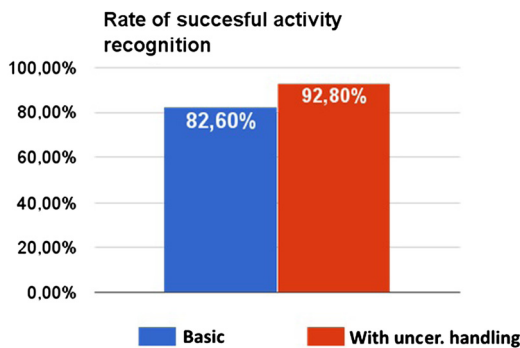


Fig. 9. Success rate of the activity recognition process under uncertainty. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

achieved gain on an activity recognizer [25] using the acquired context knowledge in an uncertain environment. We considered a simple scenario of a user that is working at home, in his/her office, with various uncertain sensors in the house. Simulated sensors provided uncertain data directly formatted as event and provided to our perception tool. Our solution provides a context knowledge that models the uncertainty, which supposedly makes the following activity recognition process more accurate. Results can be found in Fig. 9, where the blue column represents the activity recognition rate with a basic context acquisition, while the red one shows the results with our uncertainty aware solution.

3.2. Cognition

To evaluate our improvement for vision based activity recognition, we conducted experiments on several test subjects for three scenarios. Three activities were considered: remote controlling a TV, opening a door, and phoning. A test run consists of a person doing a gesture in front of a Nao robot as showed in Fig. 10. The robot uses its vision to identify the activity. It then adjusts the results based on a context knowledge provided by the smart environment. We conducted our tests on 12 subjects. Each one was asked to repeat 10 times each gesture: each scenario had 120 runs. We then compared the result of the pure vision-based recognition and the enhanced one. Results are displayed in Fig. 11. The blue column shows the pure vision-based correct recognition rate while the red one shows the improved results.

3.3. Action

DHTN was implemented and tested using both simulation and a physical robot. Some proof-of-concept tests were conducted with Nao, as shown in Fig. 12 that depicts the reactivity of DHTN.



Fig. 10. Picture of an activity recognition test run.

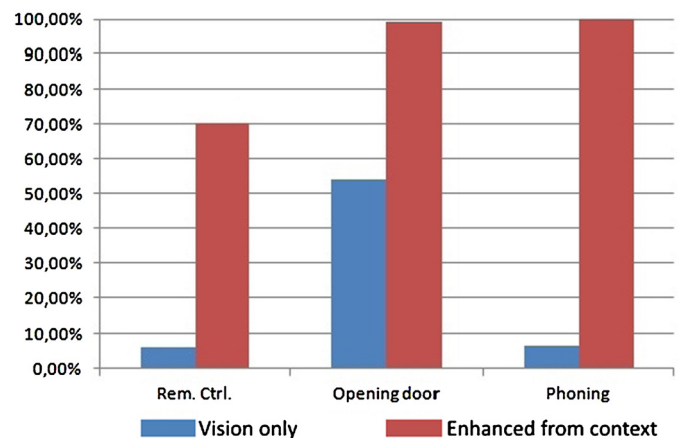


Fig. 11. Success rate of the activity recognition processes. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

In order to evaluate DHTN, we used Freedomotic and designed a simulation environment that imitates a dynamic environment. We considered a delivery scenario where the robot is asked to fetch and deliver an object to user in a variable context. For example, the simulation can change the position of the user in the house during the execution. We measured the number of tasks required to reach an objective for various levels of dynamism and compared DHTN to HTN. Results are displayed in Fig. 13.

4. Discussion

In this section, we discuss the results obtained for each layer and their synthesis with regard of the whole framework.

4.1. Perception

The perception layer was tested through simulation and measured the improvement achieved on an activity recognition process. Results, depicted in Fig. 9, show that our approach enables 10.20% more accurate activity recognition, reaching 92.80% of correctness. In fact, by providing fuzzy semantic complex events and not excluding values believed distrustful, the activity recognition can achieve better results by overlapping multiple information, in this case, for instance, location and stance. These measures prove that our solution can improve the efficiency of smart home applications.

4.2. Cognition

The evaluation conducted to assess the interest of combining vision-based activity recognition with a smart environment led to

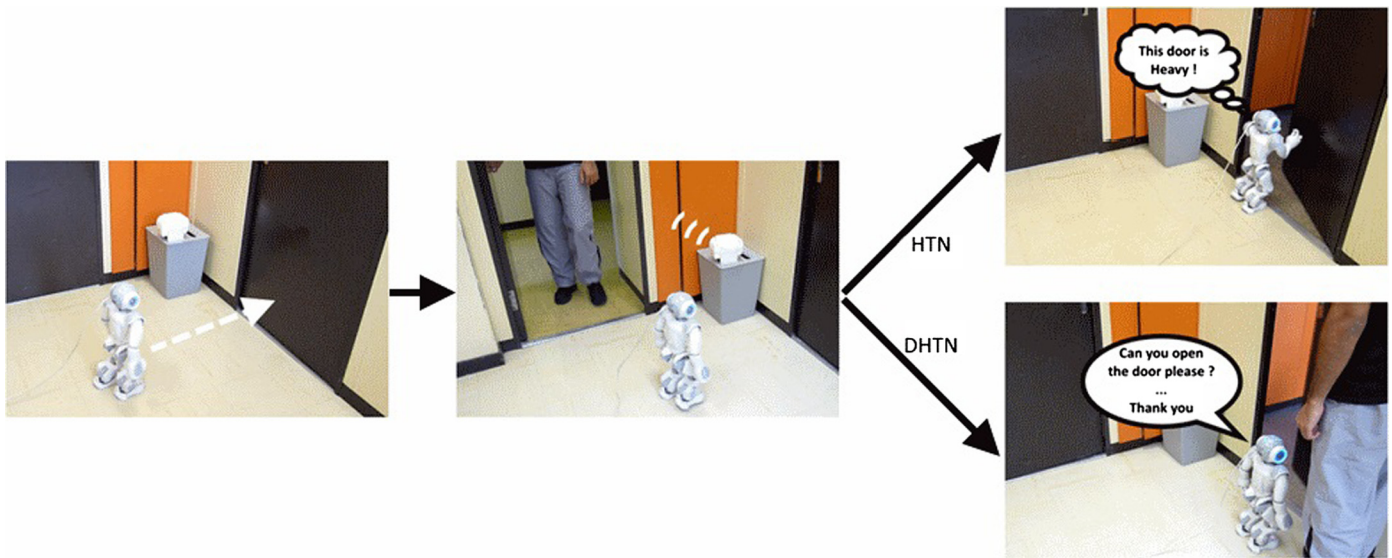


Fig. 12. Tests of task planners with a Nao robot.

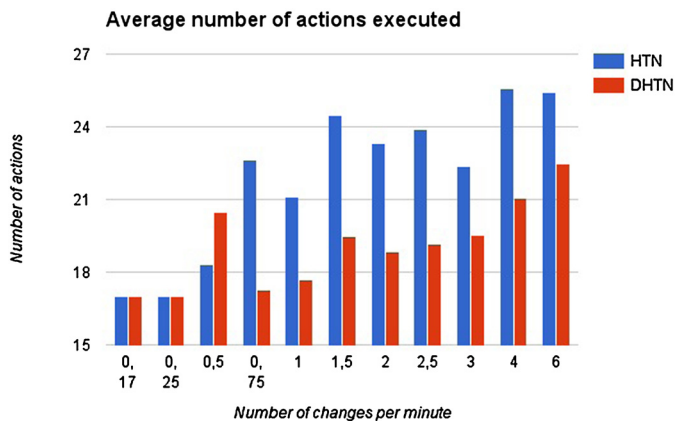


Fig. 13. Number of actions required to reach a goal for various change frequency. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

the results depicted in Fig. 11. Raw vision-based activity recognition had mediocre results: the robot was successfully able to recognize the remote control activity in 7% of the cases, 53% for the opening of the door, and 8% for the phoning activity. These poor results are explained by confusion, in fact, in most cases, the actual activity had a decent recognition confidence, but was surpassed by another, wrong, activity. On the one hand, refined results were very good: the refined “remote controlling” scenario achieved 71% rate while two others were almost reaching 100%. It proves the combination to be efficient. However, we may assert that, in some cases, the information provided by the smart environment are enough to infer the activity, without a vision process. Yet, sensors of the smart environments cannot cover all possible cases and the robot then needs its vision process. In consequence, instead of combining vision-based system and smart environments, it could be more relevant to select the best method according to the context. Yet, this way is to be explored.

4.3. Action

DHTN was compared to HTN through simulation in order to measure the gain achieved toward reaching a given goal. According the dynamism of the environment, the number of tasks performed before reaching the goal are shown in Fig. 13. As we can

notice, when using HTN, whenever the environment is dynamic, after 0.5 changes per minute, the robot has to perform more than 20 tasks to achieve the goal. This is due to fact that it does not take into account the changes, thus, it waits to encounter a failure before generating a new plan. DHTN on the other hand, by adapting the task plan to the context changes, requires much less tasks to achieves. In fact, DHTN enables a gain from 12% (6 changes per minute) to 23% (0.75 changes per minute) tasks. In other words, DHTN actually allows the robot to reach its goal quicker in dynamic environment. An artifact is observed for 0.5 changes per minute, this is explained by a change usually occurring right before the execution of one task, leading DHTN to adapt inappropriately.

4.4. Synthesis

By testing each part of the framework independently, we have shown that each of them is able to cope with the challenge they are facing by themselves, thus pointing out their role in the whole process. In fact, thanks to an uncertainty management during perception and combination between robot vision and sensors' data, the robot is able to have a less faulty activity recognition. This allows the robot and/or the medical staff to have a clear understanding and to provide an adapted health care. For taking care of the user, the robot can rely on its planner DHTN, that proves to be more efficient and quick than classical planning solution, thus, the robot can perform essential task, such as medication delivery, with a low chance of failure. Although each component of the framework were tested and proved to be efficient, the whole system needs to be tested in a real configuration, which is out of the scope of this article.

5. Conclusion

In this article, we presented a framework for robots operating in smart homes for healthcare domestic applications. We tackled multiple challenges including uncertainty of data and dynamism of the environment to provide an accurate and efficient robotic system. All our tools were tested and validated using both simulation and real case experiments. In future works, we aim to improve the decision making process to handle efficiently uncertainty and priorities. We also want to test our whole framework in a real case application using the Evident platform (<https://evident.telecom-sudparis.eu/>).

Informed consent and patient details

The authors declare that this report does not contain any personal information that could lead to the identification of the patient(s).

The authors declare that they obtained a written informed consent from the patients and/or volunteers included in the article. The authors also confirm that the personal details of the patients and/or volunteers have been removed.

Disclosure of interest

The authors declare that they have no known competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

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Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship.

References

- [1] Bevilacqua R, Felici E, Marcellini F, Glende S, Klemcke S, Conrad I, et al. Robot-era project: preliminary results on the system usability. In: International conference of design, user experience, and usability. Springer; 2015. p. 553–61.
- [2] Schroeter C, Mueller S, Volkhardt M, Einhorn E, Huijnen C, van den Heuvel H, et al. Realization and user evaluation of a companion robot for people with mild cognitive impairments. In: 2013 IEEE international conference on robotics and automation. IEEE; 2013. p. 1153–9.
- [3] Ye J, Dobson S, McKeever S. Situation identification techniques in pervasive computing: a review. *Pervasive Mob Comput* 2012;8(1):36–66.
- [4] D'Aniello G, Loia V, Orciuoli F. A multi-agent fuzzy consensus model in a situation awareness framework. *Appl Soft Comput* 2015;30:430–40.
- [5] Rodríguez ND, Cuéllar MP, Lilius J, Calvo-Flores MD. A fuzzy ontology for semantic modelling and recognition of human behaviour. *Knowl-Based Syst* 2014;66:46–60.
- [6] Okeyo G, Chen L, Wang H. Combining ontological and temporal formalisms for composite activity modelling and recognition in smart homes. *Future Gener Comput Syst* 2014;39:29–43.
- [7] Weser M, Off D, Zhang J. HTN robot planning in partially observable dynamic environments. In: 2010 IEEE international conference on robotics and automation. IEEE; 2010. p. 1505–10.
- [8] Hanheide M, Göbelbecker M, Horn GS, Pronobis A, Sjöö K, Aydemir A, et al. Robot task planning and explanation in open and uncertain worlds. *Artif Intell* 2017;247:119–50.
- [9] El-Yacoubi MA, He H, Roualdes F, Selmi M, Hariz M, Gillet F. Vision-based recognition of activities by a humanoid robot. *Int J Adv Robot Syst* 2015;12(12):179.
- [10] Selmi M, El-Yacoubi MA, Dorizzi B. Two-layer discriminative model for human activity recognition. *IET Comput Vis* 2016;10(4):273–8.
- [11] Fikes RE, Nilsson NJ. Strips: a new approach to the application of theorem proving to problem solving. *Artif Intell* 1971;2(3–4):189–208.
- [12] Tate A. Project planning using a hierarchic non-linear planner. Department of Artificial Intelligence, University of Edinburgh; 1976.
- [13] Sacerdoti ED. Planning in a hierarchy of abstraction spaces. *Artif Intell* 1974;5(2):115–35.
- [14] Georgievski I, Aiello M. An overview of hierarchical task network planning. *arXiv preprint, arXiv:1403.7426*.
- [15] Jarraya A, Ramoly N, Bouzeghoub A, Arour K, Borgi A, Finance B. Fscap: a new model for context perception in smart homes. In: OTM confederated international conferences “On the move to meaningful internet systems”. Springer; 2016. p. 465–84.
- [16] Bernhardt T, Vasseur A. Esper: event stream processing and correlation. *ONJava*. <http://www.onjava.com/lpt/a/6955>. O'Reilly.
- [17] Sfar H, Ramoly N, Bouzeghoub A, Finance B. Caredas: context and activity recognition enabling detection of anomalous situation. In: Conference on artificial intelligence in medicine in Europe. Springer; 2017. p. 24–36.
- [18] Nau DS, Au T-C, Ilghami O, Kuter U, Murdock JW, Wu D, et al. Shop2: an HTN planning system. *J Artif Intell Res* 2003;20:379–404.
- [19] Georgievski I, Aiello M. HTN planning: overview, comparison, and beyond. *Artif Intell* 2015;222:124–56.
- [20] Lallement R, De Silva L, Alami R. HATAP: an HTN planner for robotics. *arXiv preprint, arXiv:1405.5345*.
- [21] Milliez G, Lallement R, Fiore M, Alami R. Using human knowledge awareness to adapt collaborative plan generation, explanation and monitoring. In: The eleventh ACM/IEEE international conference on human robot interaction. IEEE Press; 2016. p. 43–50.
- [22] Ramoly N, Bouzeghoub A, Finance B. Context-aware planning by refinement for personal robots in smart homes. In: Proceedings of ISR 2016: 47th international symposium on robotics. VDE; 2016. p. 1–8.
- [23] Gouaillier D, Hugel V, Blazevic P, Kilner C, Monceaux J, Lafourcade P, et al. Mechatronic design of NAO humanoid. In: IEEE international conference on robotics and automation, 2009. ICRA'09. IEEE; 2009. p. 769–74.
- [24] Quigley M, Conley K, Gerkey B, Faust J, Foote T, Leibs J, et al. Ros: an open-source robot operating system. In: ICRA workshop on open source software, Kobe, Japan; 2009. p. 5.
- [25] Sfar H, Bouzeghoub A, Ramoly N, Boudy J. AGACY monitoring: a hybrid model for activity recognition and uncertainty handling. In: European semantic web conference. Springer; 2017. p. 254–69.